

Algebra 1
Unit 5
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

For questions 1–2, solve each system of equations using substitution.

1.
$$\begin{cases} y = 9x - 4 \\ y = 6x + 2 \end{cases}$$

$$\begin{aligned} 6x + 2 &= 9x - 4 \\ 2 &= 3x - 4 \\ 6 &= 3x \\ 2 &= x \end{aligned}$$
$$\begin{aligned} y &= 9(2) - 4 \\ y &= 18 - 4 \\ y &= 14 \end{aligned}$$

(2,14)

2.
$$\begin{cases} 3x - y = 4 \\ y = \frac{1}{3}x + 4 \end{cases}$$

$$\begin{aligned} 3x - \left(\frac{1}{3}x + 4\right) &= 4 \\ 3x - \frac{1}{3}x - 4 &= 4 \\ \frac{8}{3}x - 4 &= 4 \\ \frac{8}{3}x &= 8 \\ x &= 3 \end{aligned}$$
$$\begin{aligned} y &= \frac{1}{3}(3) + 4 \\ y &= 1 + 4 \\ y &= 5 \end{aligned}$$

(3,5)

Alexander was looking for new jobs. He got two different job offers. Job A offers Alexander an hourly position making \$18 an hour plus a bonus of \$750 a year. Job B offers Alexander an annual salary of \$47,000. The system of equations $\begin{cases} s = 18h + 750 \\ s = 47,000 \end{cases}$ represents the situation, where s is the salary Alexander will earn and h is the number of hours worked.

3. Solve the system of equations using substitution.

$$\begin{aligned} 47,000 &= 18h + 750 \\ 46,250 &= 18h \\ 2,569.\bar{4} &= h \end{aligned}$$

(2569. $\bar{4}$, 47000)

4. Interpret the solution to the system.
- If Alexander works 2,569. $\bar{4}$ hours at Job A, he will make the same salary as Job B, \$47,000.

Yazmin is using substitution to solve the system of equations $\begin{cases} y = 7x + 10 \\ x = -y + 2 \end{cases}$. Yazmin's partial work is shown below.

$$\begin{cases} y = 7x + 10 \\ x = -y + 2 \end{cases}$$

$$\begin{aligned} y &= 7(-y + 2) + 10 \\ y &= -7y + 2 + 10 \\ y &= -7y + 12 \\ 8y &= 12 \\ y &= \frac{12}{8} \end{aligned}$$

5. Yazmin made an error when solving for y . Circle the step where the mistake was made.

6. Find the correct solution for y .

$$\begin{aligned} y &= -7y + 14 + 10 \\ y &= -7y + 24 \\ 8y &= 24 \\ y &= 3 \end{aligned}$$

7. Using the corrected y - value, find the solution to the system.

$$\begin{aligned} x &= -3 + 2 & 3 &= 7x + 10 \\ x &= -1 & \text{or} & -7 = 7x & (-1, 3) \\ & & & -1 = x \end{aligned}$$

8. Yazmin substituted the second equation into the first to find y . Solve the system by substituting the first equation into the second to find x .

$$\begin{aligned} x &= -(7x + 10) + 2 & y &= 7(-1) + 10 \\ x &= -7x - 10 + 2 & y &= -7 + 10 \\ 8x &= -8 & y &= 3 & (-1, 3) \\ x &= -1 \end{aligned}$$

9. Did you get the same solution using both methods?

Yes

10. Solve the system of equations $\begin{cases} 2x - 4y = 8 \\ y = -\frac{3}{4}x + 3 \end{cases}$ by using substitution.

$$\begin{aligned} 2x - 4\left(-\frac{3}{4}x + 3\right) &= 8 & y &= -\frac{3}{4}(4) + 3 \\ 2x + 3x - 12 &= 8 & y &= -3 + 3 & (4, 0) \\ 5x - 12 &= 8 \\ 5x &= 20 & x &= 4 \end{aligned}$$